

Customer Behavior Analysis

Project Report

1. Project Overview

The Customer Behavior Analysis project analyzes customer shopping patterns using Python, SQL, PostgreSQL, and Power BI. The project focuses on identifying customer trends, subscription behavior, revenue contribution, and purchasing patterns to generate useful business insights for better decision-making.

2. Project Objectives

- Analyze customer purchasing behavior.
- Identify repeat buyers.
- Study subscription patterns.
- Find revenue contribution by age group.
- Understand customer preferences.

3. Tools & Technologies Used

- Python
- SQL
- PostgreSQL
- Power BI
- Pandas

4. Dataset Description

Dataset Name: customer_shopping_behavior.csv

Source: Kaggle

Rows -3900

Columns-18

Important columns include Customer ID, Age, Gender, Category, Purchase Amount, Subscription Status, Shipping Type, Review Rating, Previous Purchases, and Frequency of Purchases.

Missing data: 37 values in Review Rating column

5. Methodology

1. Collected dataset from Kaggle.
2. Performed data cleaning using Python and Pandas.
3. Handled missing values and standardized column names.
4. Connected Python with PostgreSQL database.
5. Executed SQL queries for customer behavior analysis.
6. Created interactive Power BI dashboard for visualization.

6. Python Data Preprocessing

- Python was used to clean and preprocess the dataset.
- Missing review ratings were replaced using median values by category.
- Column names were standardized and new columns such as age groups and purchase frequency days were created.
- The cleaned dataset was then loaded into PostgreSQL for SQL analysis.

7. SQL Query Analysis

Several SQL queries were executed to analyze customer purchasing behavior, subscription trends, repeat buyers, and product performance. Here are some of them:

Q. What is the total revenue generated by male vs. female customers?

```
select gender , SUM(purchase_amount) as revenue
from customer
group by gender;
```

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Q. Which are the top 5 products with the highest average review rating?

```
select item_purchased, round(AVG(review_rating::numeric),2) as
"Average Product Rating"
from customer
group by item_purchased
order by avg(review_rating) desc
limit 5;
```

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Q. Which 5 products have the highest percentage of purchases with discounts applied?

```
select item_purchased,
round(100*SUM(CASE when discount_applied ='Yes' then 1 else 0
end)/count(*),2) as discount_rate
from customer
group by item_purchased
order by discount_rate desc
limit 5;
```

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

Q. Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

```
with customer_type as (
select customer_id, previous_purchases,
case
    when previous_purchases = 1 then 'New'
    when previous_purchases between 2 and 10 then 'Returning'
    else 'Loyal'
end as customer_segment
from customer
)
```

	customer_segment text	Number of Customer bigint
1	Loyal	3116
2	New	83
3	Returning	701

```
select customer_segment, count(*) as "Number of Customer"
from customer_type
group by customer_segment;
```

8. Power BI Dashboard

The Power BI dashboard provides a visual overview of customer behavior insights. It includes KPIs such as total customers, average purchase amount, and average review rating. The dashboard also visualizes revenue by category, subscription status, and sales by age group.



9. Key Findings

- Clothing category generated the highest revenue.
- Subscribed customers contributed significant revenue.
- Young adults had the highest sales contribution.
- Discount strategies influenced customer purchasing behavior.
- Repeat buyers were more likely to subscribe.

10. Conclusion

The Customer Behavior Analysis project successfully identified important customer trends and purchasing patterns using SQL and Power BI. The project demonstrates how data analysis techniques can help businesses improve customer understanding, marketing strategies, and decision-making processes.